

PROPER PURELY 1-UNRECTIFIABLE METRIC SPACES HAVE LOCALLY FLAT UNIFORM SEPARATION

ARXIV_AUTOMATIC_MATH RUN FA_BANACH.001

STATUS

Result type: partial result. Status: likely valid pending human review.

This packet addresses Question 2.7 (TeX label `q:spu p1u`) in Aliaga's exposition [1]. The question asks whether every purely 1-unrectifiable metric space M has $\text{lip}(M)$ separating points uniformly. The packet proves an affirmative answer for every proper purely 1-unrectifiable metric space. Here proper means that every closed bounded ball is compact. This is still not the full arbitrary metric-space case.

OPEN-PROBLEM EVIDENCE

The source question appears on page 6 of the rendered source PDF:

This ends the proof. □

Whether the converse of Proposition 2.6 holds is, to the best of our knowledge, an open question at the time of writing.

Question 2.7. *Let M be a purely 1-unrectifiable metric space. Does $\text{lip}(M)$ separate points of M uniformly?*

Transcription: Let M be a purely 1-unrectifiable metric space. Does $\text{lip}(M)$ separate points of M uniformly?

DEFINITIONS

A metric space is proper if every closed bounded ball is compact. A real-valued Lipschitz function f on M is locally flat at $p \in M$ if its asymptotic Lipschitz constant at p is zero; equivalently, the small-scale Lipschitz constants of f at p tend to zero. We write $\text{lip}(M)$ for the space of Lipschitz functions that are locally flat at every point. The space $\text{lip}(M)$ separates points uniformly if for every $a, b \in M$ and every $\varepsilon > 0$ there is $h \in \text{lip}(M)$ with $\text{Lip}(h) \leq 1$ and

$$h(b) - h(a) > d(a, b) - \varepsilon.$$

INPUT FROM THE COMPACT THEORY

We use the compact approximation lemma quoted in [1, Lemma 2.11] from Bate's work [2]. In the form needed here, if K is compact and purely 1-unrectifiable, $f \in \text{Lip}(K)$, and $\eta > 0$, then there are $g \in \text{Lip}(K)$ and $\delta > 0$ such that

- (i) $|g(z) - f(z)| \leq \eta$ for all $z \in K$;
- (ii) for all $u \neq v$ in K ,

$$\frac{|g(u) - g(v)|}{d(u, v)} \leq \frac{|f(u) - f(v)|}{d(u, v)} + \eta;$$

(iii) for all $u \neq v$ in K with $d(u, v) \leq \delta$,

$$\frac{|g(u) - g(v)|}{d(u, v)} \leq \eta.$$

PROOF INTUITION

The compact AGPP theorem is proved by repeatedly applying the approximation lemma until small-scale slopes are forced to zero. For a proper noncompact space, closed balls are compact, so we can run the same idea on larger and larger balls. After improving a function on a compact ball, extend it back to the whole space by McShane. Later improvements on larger balls may change the old ball, but the lemma's slope inequality says they can increase old slopes only by the chosen error. Choosing summable errors with tails tending to zero makes the final diagonal limit locally flat at every point, because every point eventually sits in the interior of some compact ball.

MAIN THEOREM

Theorem 1. *Let M be a proper purely 1-unrectifiable metric space. Then $\text{lip}(M)$ separates points of M uniformly.*

Proof. Fix $a, b \in M$ and $\varepsilon > 0$. The case $a = b$ is trivial, so put $D = d(a, b) > 0$. We will construct a locally flat function whose Lipschitz constant is slightly larger than 1, then rescale.

Choose a summable sequence $(\eta_n)_{n \geq 1}$ of positive numbers so small that

$$S := \sum_{n=1}^{\infty} \eta_n < \frac{\varepsilon}{D+3}.$$

Let $o = a$ and choose radii $R_n \uparrow \infty$ with $b \in K_1$, where

$$K_n = \overline{B}(o, R_n).$$

Each K_n is compact by properness and purely 1-unrectifiable as a subset of M .

Set $F_0(z) = d(a, z)$ on M . Inductively suppose that F_{n-1} is defined on M . Apply the compact approximation lemma to $F_{n-1}|_{K_n} \in \text{Lip}(K_n)$ with parameter η_n . This gives $g_n \in \text{Lip}(K_n)$ and $\delta_n > 0$ such that

$$\begin{aligned} (1) \quad & |g_n(z) - F_{n-1}(z)| \leq \eta_n & (z \in K_n), \\ (2) \quad & \frac{|g_n(u) - g_n(v)|}{d(u, v)} \leq \frac{|F_{n-1}(u) - F_{n-1}(v)|}{d(u, v)} + \eta_n & (u \neq v \in K_n), \\ (3) \quad & \frac{|g_n(u) - g_n(v)|}{d(u, v)} \leq \eta_n & (u \neq v \in K_n, d(u, v) \leq \delta_n). \end{aligned}$$

Let F_n be any McShane extension of g_n to M . Then $F_n|_{K_n} = g_n$ and

$$\text{Lip}(F_n) \leq \text{Lip}(g_n) \leq \text{Lip}(F_{n-1}) + \eta_n.$$

Consequently $\text{Lip}(F_n) \leq 1 + \sum_{k=1}^n \eta_k$.

For a fixed N , the restrictions $F_n|_{K_N}$ are uniformly Cauchy for $n \geq N$. Indeed, if $n > N$, then $K_N \subset K_n$, and (1) gives

$$\sup_{z \in K_N} |F_n(z) - F_{n-1}(z)| \leq \eta_n.$$

Thus the restrictions converge uniformly on every K_N to a compatible function $F : M \rightarrow \mathbb{R}$. The Lipschitz estimates pass to the limit and give

$$\text{Lip}(F) \leq 1 + S.$$

Since $a, b \in K_1$, telescoping (1) also gives

$$|F(a) - F_0(a)| \leq S, \quad |F(b) - F_0(b)| \leq S.$$

Therefore

$$F(b) - F(a) \geq D - 2S.$$

We now prove that F is locally flat. Fix $p \in M$ and $\alpha > 0$. Choose N so large that p lies in the interior of K_N and

$$\sum_{n=N}^{\infty} \eta_n < \alpha.$$

Let $\rho > 0$ be such that $B(p, \rho) \subset K_N$. For $u, v \in K_N$ with $u \neq v$ and $d(u, v) \leq \delta_N$, (3) gives the slope bound η_N for F_N . For every $m > N$, applying (2) on K_m to the same pair gives

$$\frac{|F_m(u) - F_m(v)|}{d(u, v)} \leq \frac{|F_{m-1}(u) - F_{m-1}(v)|}{d(u, v)} + \eta_m.$$

Letting $m \rightarrow \infty$ yields

$$\frac{|F(u) - F(v)|}{d(u, v)} \leq \sum_{n=N}^{\infty} \eta_n < \alpha$$

whenever $u, v \in K_N$ and $d(u, v) \leq \delta_N$. In particular this holds for $u, v \in B(p, \rho/2)$ with $d(u, v) \leq \delta_N$. Hence the asymptotic Lipschitz constant of F at p is 0. Since p was arbitrary, $F \in \text{lip}(M)$.

Finally set $H = F/(1 + S)$. Then $H \in \text{lip}(M)$ and $\text{Lip}(H) \leq 1$. Moreover

$$H(b) - H(a) \geq \frac{D - 2S}{1 + S} = D - \frac{S(D + 2)}{1 + S} > D - \varepsilon,$$

by the choice of S . Thus $\text{lip}(M)$ separates the pair a, b uniformly, and the pair was arbitrary. $\square$

VERIFICATION NOTES

The proof is non-computational. The main things to check are:

- The compact approximation lemma is applied only to compact balls, which are purely 1-unrectifiable because pure unrectifiability passes to subsets.
- McShane extension is used only after the compact modification; all later estimates on older compact balls are made before extension matters, because the next compact ball contains the old one.
- Local flatness follows from the summable tail of the slope errors and from the fact that every point eventually lies in the interior of a compact ball in the exhaustion.

NOVELTY AND LIMITATIONS

The run indexes were searched for 2606.02918, `q:spu, locally flat, separates points uniformly, proper, boundedly compact, and purely 1-unrectifiable`. The only same-question packet found was the earlier metric-wedge partial result from this run. A bounded web phrase search on 2026-06-29 for proper/boundedly compact variants of the locally-flat separation question found no exact proper-space closure result.

This is not a full answer to Question 2.7. The proof uses compact neighborhoods essentially: without local compactness, the exhaustion does not place small neighborhoods of a point inside a compact pure set, and the argument gives no local-flatness control.

HUMAN REVIEW RECOMMENDATION

Reviewers should focus on the diagonal/exhaustion step: the restrictions $F_n|_{K_N}$ are Cauchy, and the slope estimates on K_N survive all later steps by the one-sided slope inequality in Bate's compact lemma. If that bookkeeping is accepted, the theorem is a direct noncompact extension of the compact approximation proof.

REFERENCES

- [1] R. J. Aliaga, *Lipschitz-free spaces and purely 1-unrectifiable metric spaces*, arXiv:2606.02918, 2026. <https://arxiv.org/abs/2606.02918>
- [2] D. Bate, *Purely unrectifiable metric spaces and perturbations of Lipschitz functions*, Acta Math. **224** (2020), 1–65.
- [3] R. J. Aliaga, C. Gartland, C. Petitjean, and A. Prochazka, *Purely 1-unrectifiable metric spaces and locally flat Lipschitz functions*, Trans. Amer. Math. Soc. **375** (2022), 3529–3567. <https://arxiv.org/abs/2103.09370>